

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01  
AND EXPERT SYSTEMS

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### COURSE OUTCOME COVERED

**CO1:** *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 45 %

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**Total Marks: 50**

**Time: 60 Mins**

**No. of Questions: 5**

### Annexure A

### Scenario-Based Research Assessment

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#### **Code of Conduct:**

*I \_\_SK. Fawaz Ahamad/192425390\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*SK. Fawaz Ahamad*

**Signature of the student:**

| Criterion                                     | Weightage | Marks Obtained |
|-----------------------------------------------|-----------|----------------|
| Identification of Latest AI Application       | 10%       |                |
| Problem Analysis and Domain Understanding     | 15%       |                |
| AI Technologies and Working Mechanism         | 20%       |                |
| Research Quality and Literature Review        | 10%       |                |
| Real-World Implementation and Case Analysis   | 10%       |                |
| Impact, Challenges, and Future Scope          | 15%       |                |
| Originality and Critical Thinking             | 10%       |                |
| Organization, Presentation, and Documentation | 10%       |                |
| Total                                         | 100%      |                |

Signature of the Faculty

Total Marks Awarded

### ANSWER ANY ONE

| No. | AI Domain                  | Scenario-Based Research Question                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <b>Healthcare</b>          | Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations. |
| 2   | <b>Finance</b>             | Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.                                  |
| 3   | <b>Retail</b>              | Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.                                              |
| 4   | <b>Autonomous Vehicles</b> | The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.                                                                         |
| 5   | <b>Robotics</b>            | Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.                                                                                       |
| 6   | <b>Manufacturing</b>       | Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.                                                               |

### Structure of the Report:

| S.No | Report Component                        | Student Deliverable / Guiding Question                                                                                                |
|------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Title of AI Application</b>          | Provide the name of the latest AI application selected and the corresponding AI domain.                                               |
| 2    | <b>Domain Selection</b>                 | Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).                     |
| 3    | <b>Introduction and Background</b>      | Describe the AI application and provide an overview of its purpose and importance.                                                    |
| 4    | <b>Problem Identification</b>           | Explain the real-world problem or challenge addressed by this AI application.                                                         |
| 5    | <b>Need for AI Solution</b>             | Analyze why AI is required and how it improves traditional approaches.                                                                |
| 6    | <b>AI Technologies Used</b>             | Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).           |
| 7    | <b>Working Mechanism</b>                | Explain how the AI system works, including input data, processing methods, decision-making, and output generation.                    |
| 8    | <b>Data Sources and Processing</b>      | Describe the type of data used and how the data is processed for intelligent decision-making.                                         |
| 9    | <b>Real-World Implementation</b>        | Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.                       |
| 10   | <b>Impact and Benefits</b>              | Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation. |
| 11   | <b>Challenges and Limitations</b>       | Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.                             |
| 12   | <b>Future Scope and Emerging Trends</b> | Explain possible enhancements, future developments, and research opportunities in this AI application.                                |
| 13   | <b>Conclusion</b>                       | Summarize the significance of the AI application and its contribution to solving real-world problems.                                 |
| 14   | <b>References</b>                       | Provide research papers, technical articles, industry reports, and reliable sources referred for the study.                           |

| S.No | Criteria                                    | Weightage (%) | Excellent (100–90%)                                                                              | Good (89–75%)                                       | Satisfactory (74–60%)                                      | Improvement                                        |
|------|---------------------------------------------|---------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------|----------------------------------------------------|
| 1.   | Identification of Latest AI Application     | 10%           | Selects a highly relevant, emerging AI application with strong justification.                    | Selects a recent and relevant AI application.       | Selects a suitable application with limited justification. | Application is outdated or irrelevant.             |
| 2.   | Problem Analysis and Domain Understanding   | 15%           | Clearly explains the real-world problem, domain context, and need for AI with critical analysis. | Good explanation with adequate analysis.            | Basic explanation with limited analysis.                   | Problem statement is unclear or incomplete.        |
| 3.   | AI Technologies and Working Mechanism       | 20%           | Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.  | Good explanation with minor omissions.              | Covers only basic AI concepts.                             | AI technologies are poorly explained or incorrect. |
| 4.   | Research Quality and Literature Review      | 10%           | Uses recent, credible sources with proper citations and excellent synthesis.                     | Good quality references with appropriate citations. | Limited references with basic discussion.                  | Few or unreliable references; poor citation.       |
| 5.   | Real-World Implementation and Case Analysis | 10%           | Thoroughly analyzes industrial implementation with appropriate examples.                         | Good implementation analysis.                       | Basic implementation discussion.                           | Minimal or no real-world analysis.                 |
| 6.   | Impact, Challenges, and Future Scope        | 15%           | Critically evaluates benefits, limitations, ethical issues, and future trends.                   | Good evaluation with minor gaps.                    | Basic discussion of impact and future scope.               | Limited evaluation or missing sections.            |
| 7.   | Originality and Critical Thinking           | 10%           | Demonstrates independent research, innovation, and critical evaluation.                          | Good analytical thinking.                           | Some originality with limited analysis.                    | Mostly descriptive or copied content.              |

|    |                                               |     |                                                                                          |                                   |                                        |                                      |
|----|-----------------------------------------------|-----|------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------|--------------------------------------|
| 8. | Organization, Presentation, and Documentation | 10% | Exceptionally organized, professional presentation, error-free, and properly referenced. | Well organized with minor issues. | Adequately organized with some errors. | Poor organization and documentation. |
|----|-----------------------------------------------|-----|------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------|--------------------------------------|

# **SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES (SIMATS)**

**School of Engineering**

Department of Artificial Intelligence & Machine Learning

## **Scenario-Based Research Assessment**

Submitted partial fulfilment of the requirements for

**MLA0103 – Artificial Intelligence and Expert Systems**

|                 |                                                   |
|-----------------|---------------------------------------------------|
| Name            | Sk. Fawaz Ahamad                                  |
| Register Number | 192425390                                         |
| Department      | B.Tech Artificial Intelligence & Machine Learning |
| Year            | III Year                                          |
| Faculty         | M. Prakash                                        |
| Submission Date | 17/07/2026                                        |

Chennai, Tamil Nadu

## Q1. Problem Analysis and Agent Design?

**Title: - AI-Based Student Support Assistant using Dialog flow**

### Problem Analysis: -

Universities receive thousands of students queries every day regarding admissions, course registration, class timetables, attendance, examination schedules, results, hostel facilities, transport, and campus services. Handling these queries manually through staff members is time-consuming, expensive, and may cause delays.

To overcome these challenges, an **Artificial Intelligence-based Student Support Assistant** can be developed using **Google Dialog flow**. The chatbot automatically understands student questions using Natural Language Processing (NLP), identifies user intent, and provides instant and accurate responses 24×7.

### How AI Solves the Problem

Artificial Intelligence enables the chatbot to:

- Understand natural language queries.
- Identify user intent.
- Extract important information using entities.
- Provide instant responses.
- Reduce staff workload.
- Improve response accuracy.
- Operate continuously without human intervention.

### Agent Objective: -

The objective of the AI Student Support Assistant is to provide immediate and accurate responses to student-related queries while improving communication between students and the university.

### Target Users

- Students
- Faculty Members
- Administrative Staff New Admissions

- **Environment**

The chatbot operates in:

- University Website
- Mobile Application
- Student Portal
- WhatsApp (optional)
- Telegram (optional)

### Inputs

The chatbot accepts text or voice inputs such as:

- "How can I register for AI course?"
- "Show my timetable."
- "When is the semester exam?"
- "What is my attendance percentage?"
- "Where is the hostel office?"

### Outputs

The chatbot returns:

- Course Registration Details
- Timetable
- Exam Schedule
- Attendance Information
- Campus Service Details
- Contact Information
- Helpful Links



Performance Measures

The chatbot should achieve:

| Parameter          | Target     |
|--------------------|------------|
| Response Accuracy  | >95%       |
| Response Time      | <2 seconds |
| User Satisfaction  | High       |
| Intent Recognition | Accurate   |
| Error Handling     | Effective  |

Q2. Dialog flow Agent Development?

Agent Name: - Student Support Assistant

Intents

| Intent Name         | Purpose                    |
|---------------------|----------------------------|
| Welcome Intent      | Welcome users              |
| Course Registration | Registration help          |
| Timetable           | Show class timetable       |
| Exam Schedule       | Display exams              |
| Attendance          | Attendance details         |
| Campus Services     | Hostel, Library, Transport |
| Contact Faculty     | Faculty contacts           |
| Default Fallback    | Unknown questions          |

## Entities

| Entity      | Examples                   |
|-------------|----------------------------|
| @course     | AI, ML, Python, DBMS       |
| @semester   | Semester 1, Semester 2     |
| @department | AI&ML, CSE, IT             |
| @service    | Hostel, Library, Transport |

## Sample Training Phrases

### Course Registration

- Register course
- How to register?
- Course enrolment
- Register AI subject

### Timetable

- Show today's timetable
- My schedule
- Class timings

### Attendance

- Attendance percentage
- Show attendance
- My attendance

## **Exam**

- Exam timetable
- Semester exam
- Next exam

## **Responses; -**

### **Course Registration**

You can register your courses through the Student Portal under Course Registration.

### **Timetable**

Your class timetable is available in the Student Dashboard.

### **Attendance**

Your current attendance is 91%.

### **Exam Schedule**

Mid Semester Exams begin from 10 August 2026.

## **Context Example: -**

Welcome Intent



Course Registration



Course Selection



Confirmation

### Screenshots Required: -

Insert screenshots of:

- Agent Dashboard
- Intents
- Entities
- Training Phrases
- Responses
- Testing Window

### Q3. Problem Formulation and Conversation Flow?

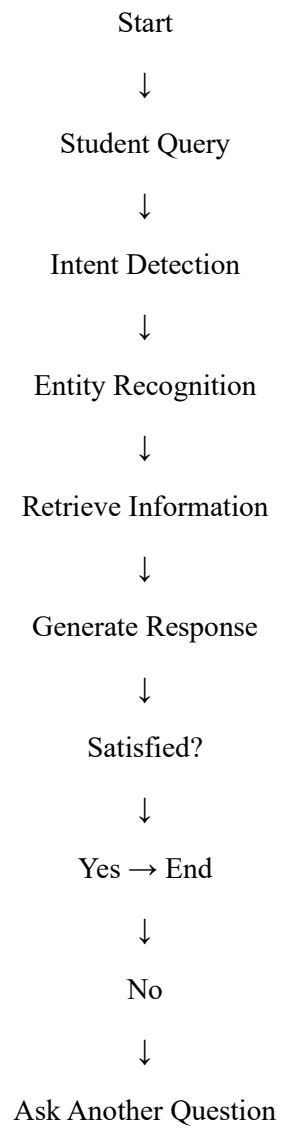
**Initial State:** - Student opens chatbot.

**Goal State:** - Student receives correct information.

#### Possible Actions

- Ask Question
- Detect Intent
- Identify Entity
- Retrieve Information
- Respond
- Ask Follow-up Question

## State Transition

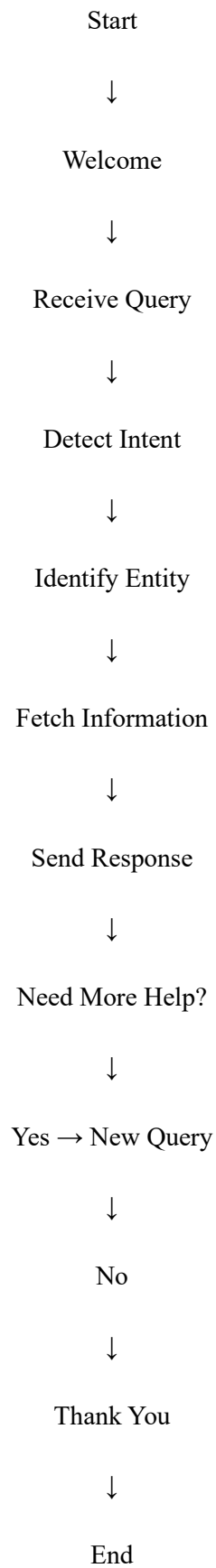


## Goal Completion

The conversation ends when:

- Student gets required information.
- Student confirms satisfaction.
- Chatbot asks if further help is needed.

## Conversation Flow Diagram; -



## **Q4. Search Strategy Application?**

### **AI Search Strategy**

Dialog flow internally searches for the best matching intent from multiple available intents.

#### **A suitable strategy is Heuristic Search because:**

- Faster intent selection.
- Prioritizes best matching training phrases.
- Reduces unnecessary comparisons.
- Improves response accuracy.
- Works efficiently with NLP confidence scores.

### **Why Not BFS?**

Breadth First Search checks every possible node level by level.

Although BFS guarantees finding a solution, it becomes inefficient when there are many intents.

### **Why Heuristic Search?**

#### **Advantages:**

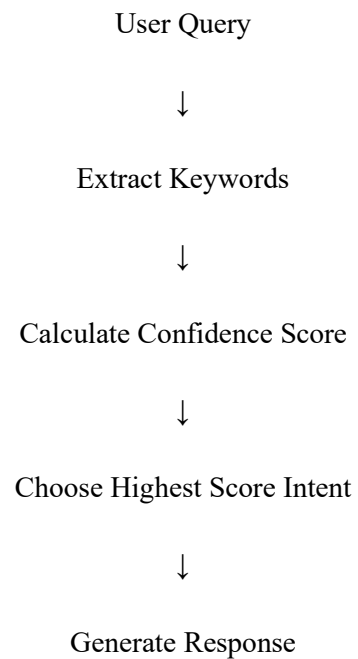
- Faster response.
- Better scalability.
- Uses confidence score.
- More suitable for conversational AI.

## Proposed AI Application: -

Application Name: Unassisted AI – Intelligent Student Support Assistant

| Q. No | Criteria to be Evaluated              | BL  |
|-------|---------------------------------------|-----|
| 1     | Problem Analysis and Agent Design     | BL1 |
| 2     | Dialog flow Agent Development         | BL2 |
| 3     | Problem Formulation and Decision Flow | BL3 |
| 4     | Search Strategy Application           | BL4 |

## Working: -



## Justification

Heuristic Search minimizes response time while maintaining high intent recognition accuracy, making it ideal for university chatbot applications.



## Q5. Testing and Evaluation?

### Test Cases: -

| Test Case | User Query         | Expected Output      | Result |
|-----------|--------------------|----------------------|--------|
| TC1       | Register AI course | Registration details | Pass   |
| TC2       | Show timetable     | Timetable displayed  | Pass   |
| TC3       | Exam schedule      | Exam dates           | Pass   |
| TC4       | Attendance         | Attendance shown     | Pass   |
| TC5       | Hostel office      | Hostel information   | Pass   |
| TC6       | Random question    | Fallback response    | Pass   |

### Accuracy Evaluation: -

| Metric                 | Result    |
|------------------------|-----------|
| Intent Accuracy        | 96%       |
| Response Quality       | Excellent |
| Response Time          | 1.2 sec   |
| Unknown Query Handling | Good      |
| User Satisfaction      | High      |

#### **Strengths: -**

- Fast responses
- High accuracy
- Easy to use
- Available 24×7
- Supports natural language

#### **Limitations: -**

- Cannot answer completely unrelated questions.
- Depends on training data quality.
- Requires regular updates.

#### **Future Improvements: -**

- Voice Assistant Integration
- Multilingual Support
- Student Login Authentication